

Tulasi Nayak

AI Engineer · LLM Applications · AI Agents · Fine-Tuning

Saarbrücken, Germany

tulasinayak0701@gmail.com

LinkedIn: [linkedin.com/in/tulasi-nayak](https://www.linkedin.com/in/tulasi-nayak) | GitHub: github.com/tulasinayak | Portfolio: tulasinayak.github.io

Profile

AI Engineer who takes LLM systems from idea to working product: **multi-agent applications**, **RAG pipelines**, **FastAPI** backends and **QLoRA fine-tuning** of open models such as Qwen and Llama. Research at DFKI and CISPA taught me what reliable AI needs most: strong baselines, careful **evaluation** and results you can trust.

Experience

Master's Thesis Student, Machine Learning / NLP

05/2025 – Present

German Research Center for Artificial Intelligence (DFKI), Saarbrücken, Germany

- **Fine-tuned Qwen3-8B and Llama-3.1-8B** with QLoRA (Hugging Face, PEFT, PyTorch) on on-premise A100 GPU servers.
- **Curated the training and evaluation data**: rebuilt the pipeline after finding 82% of labels were joined by unreliable fuzzy matching; removed 337 contaminated records.
- Improved models **iteratively**: weighted F1 **0.58 zero-shot**, **0.72 few-shot**, **0.76 fine-tuned** vs. 0.51 baseline; compared generation against a classification head.
- Built the **evaluation framework** (grouped CV, bootstrap CIs, ablations, error analysis), exposing a prompt shortcut.

Research Assistant, Machine Learning (Part-time)

08/2024 – Present

CISPA Helmholtz Center for Information Security, Saarbrücken, Germany

- **Numerical data analysis**: built XGBoost models on engineered features, pretrained embeddings and GPT-generated features, with correlation analysis.
- Turned research code into **reproducible Python workflows** with automated quality checks, together with researchers at CISPA and LMU Munich.

Projects

TRIPBuddy: Multi-Agent AI Travel Planner (Python, FastAPI, LangChain)

[GitHub](#)

- Built the **FastAPI backend** with Pydantic API contracts: an orchestrator and 7 specialist LLM agents turn one request into a structured plan with spoken output.
- A model layer switches between OpenAI, Claude and **self-hosted Ollama models** without code changes; external service failures fall back gracefully.

depscan: Agentic Vulnerability Scanner on Self-Hosted LLMs (Python, Ollama)

[GitHub](#)

- Process automation end to end: clones a repository, maps dependencies to CVEs via the **OSV.dev API**, and an **open-weight LLM** running locally judges each CVE; evaluated on a ground-truth test set.

DocuMind Agent: RAG Chat Assistant (LangChain, FAISS)

[GitHub](#)

- Built a **RAG chat application** over PDFs with embeddings and a **FAISS** vector index, grounding answers in sources.

Publications

B. S. Kamath and T. Nayak, "Construction of a 3D Model of a University Campus for Easing New Visitor Navigation Around the Campus Using Blender and Three.js," *ICEEICT 2024 (IEEE)*, Trichirappalli, India. [IEEE Xplore](#)

Education

M.Sc. Computer Science, Saarland University, Saarbrücken, Germany

10/2023 – 2026

- **Coursework**: Generative AI, Trustworthy ML. **Seminars**: Privacy in ML; Emotion Recognition in Bodily Behavior.

B.E. Information Science, NMAM Institute of Technology, Udupi, India

08/2019 – 06/2023

- **Grade**: 1.3 (German scale, "Sehr gut"). **Leadership**: ACM student council manager; led final-year project team.

Skills

LLM Applications & Agents: Multi-Agent Systems, Tool Calling, RAG, LangChain, FAISS, Prompt Engineering, OpenAI, Claude, Ollama (Qwen, Llama)

Fine-Tuning & Models: QLoRA, PEFT, Hugging Face Transformers, PyTorch, Optuna, Multi-GPU A100 Servers

Data & Evaluation: Training and Evaluation Dataset Curation, Data Pipelines, Data Quality, pandas, XGBoost, Numerical Data Analysis, Cross-Validation, Error Analysis

Backend: Python, FastAPI, REST APIs, Pydantic, SQL, Streamlit, Gradio, Bash

Infrastructure & Tools: Docker, Kubernetes (basics), Git, GitHub, Linux, Claude Code, OpenAI Codex

Languages & Interests

Languages: English C1 (business fluent) | German A2 (Goethe-Institut, ongoing) | Hindi C2 | Konkani native

Interests: Hiking, Drawing, Traveling